

MSFormer: Multi-scale Transformer for Robust Fault Diagnosis of Machines under Complex Conditions

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Abstract

In practical rotating machinery monitoring, obtaining highly discriminative fault features from complex vibration signals remains a significant challenge for deep learning based diagnostic models. In this paper, a novel intelligent fault diagnosis method named MSFormer is proposed. The MSFormer incorporates a parallel multi scale Convolutional Neural Network (CNN) architecture and hierarchical Transformer modules to comprehensively process 1D vibration signals. By utilizing varying kernel sizes, the multi scale CNN backbone extracts both high frequency local transient impulses and low frequency global degradation trends. Subsequently, the Transformer modules are employed to model the long range dependencies within the extracted feature sequences, effectively mitigating the interference of environmental noise. Extensive experiments are conducted on a standard bearing dataset to evaluate the proposed method. Four state of the art models are compared under the same experimental settings. Quantitative metrics and qualitative tools are utilized for comprehensive evaluation. Experimental results indicate that MSFormer achieves a 92.67% accuracy, 92.54% F1-score, and 93.22% precision, demonstrating significant superiority. Furthermore, ablation experiments confirm that both the multi scale CNN and Transformer components are essential for the enhanced performance and stability. MSFormer exhibits excellent intra class compactness and inter class separability, providing a powerful and precise intelligent solution for mechanical fault diagnosis.

Keywords: bearing fault diagnosis; multi scale CNN; Transformer; feature fusion

1. Introduction

The heart of contemporary industrial systems are the components of rotating machinery (especially rolling bearings). Since unforeseen structural deterioration can cause catastrophic economic damages and extreme safety risks, precise and timely smart fault detection has become a priority research topic. At the beginning of the field, conventional fault diagnosis used a large number of signal processing methods based on experience, and empirical mode decomposition and wavelet transform [1,2] were devised as methods to extract the frequencies of the defect. In order to reduce human intervention, the traditional machine learning model [3,4] was later provided to automatically determine the type of fault. In the fast development of deep learning, the use of standard convolutional neural networks (CNNs) [5,6] has been common to automatically discover discriminative features of raw monitoring data. However, non stationary vibration signals are inherent in real world industrial environments, which are dynamic and open in nature, and as such

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mechanical equipment would produce non stationary vibration signals under different operating conditions. This operational variability is a very serious exacerbation of the lack of annotated fault data, becoming a very serious challenge to the possibility of reliable diagnosis with very limited sample sizes. To specifically tackle this data scarcity challenge, recent studies have attempted to integrate domain expertise. For example, multi task dynamic learning informed by prior knowledge [33] and self supervised feature learning augmented with prior knowledge [34] have been successfully developed to improve machine fault diagnosis by a few-shots.

To address these issues, many sophisticated deep learning models have been developed. First, intelligent diagnosis systems based on deep belief networks [7] and sparse auto encoders [8] were created to learn latent characteristic representations of complex signals. Recurrent neural networks (RNNs) [9] and long short term memory (LSTM) networks [10] were used to model sequential patterns in order to track the temporal changes inherent in vibration data. In addition, attention based CNNs [11] and deep residual learning networks [12] were developed that focused on informative fault properties and avoided irrelevant variations. In the case of multi sensor systems, graph convolutional networks [13] and capsule networks [14] were proposed to increase the synthesis of spatial and temporal attributes. Unsupervised domain adaptation (UDA) techniques [15] and deep transfer learning networks [16] also tried to reduce the gap in distributions between the two domains to maintain performance in the context of variable working conditions. In particular, Zhang et al. [17] suggested an RNN based diagnostic algorithm that detects the type of faults by 1D to 2D conversion of the vibration signals of the time series, and adding Gated Recurrent Units (GRUs), although this is a required 1D to 2D conversion, which breaks the physical continuity of time series data and introduces excessive computational redundancy. To alleviate the spatial and temporal trade off, multi branch CNNs [18] and dilated convolutional networks [19] were developed to directly process raw signals. Most recently, vanilla Transformer models [20] have proven themselves to have superior abilities in parallel long range temporal dependencies. As an example, Ding et al. [21] developed a time frequency Transformer (TFT) model that is motivated by the self attention mechanism to extract high dimensional abstractions of time frequency representations (TFR) although this method is again largely reliant on complex manual TFR computations. In order to increase the cross domain robustness, adversarial domain adaptation networks [22] and feature disentanglement auto encoders [23] were investigated, to isolate fault features and operating conditions. In addition, Zhao et al. [24] suggested a semi supervised mutual assistance domain generalization (SemiDGF) network in order to achieve multidomain generalization with single source annotations. However, those domain generalization networks [25] and meta learning diagnostic schemes [26] remain strongly dependent on auxiliary data or complicated pseudo labeling, and are thus extremely vulnerable to severe overfitting or optimization failure in few shot regimes. Lastly, Chen et al. [27] suggested a bearing fault diagnosis mechanism that uses a multi scale CNN and LSTM model to automatically extract multi frequency features, the sequentiality of the LSTM limits the inference rate, and its use of signal down sampling risks the loss of important high frequency transient fault features.

In order to escape the strong dependence on multi source auxiliary datasets, time consuming time frequency preprocessing, lossy down sampling processes, and unreliable domain alignment heuristics, a new end to end intelligent fault diagnosis architecture called MSFormer is introduced in this paper. In a strictly constrained single domain setup, the MSFormer integrates a parallel multi scale 1D CNN backbone with hierarchical Transformer encoders in a complex way. In particular, the multi scale 1D CNN backbone leverages different kernel sizes to implicitly learn high frequency local transient impulses and low

frequency global degradation patterns using raw 1D vibration signals without any prior information loss. Lightweight Transformer encoders are then presented to independently model long range global dependencies in the extracted feature sequences, which effectively suppresses intricate industrial background noise. This architecture does not need to use conventional statistical fusion ensembles, or any elaborate geometric distribution alignment loss terms, and inherently learns superior domain invariant representations using only the internal feature mining motivation of the network.

In summary, the core academic contributions of this paper are outlined as follows:

- A proposed MSFormer, a novel which is end to end hybrid diagnostic architecture, organically integrating the local feature extraction advantages of multi scale 1D CNNs with the global parallel temporal modeling capabilities of Transformers to achieve robust diagnosis in complex environments.
- MSFormer completely eliminates the prerequisite for cumbersome 1D to 2D signal preprocessing and bloated domain alignment or feature decoupling mechanisms, establishing a pure and highly data efficient 1D streaming paradigm.
- Extensive comparative experiments against state of the art diagnostic models under a rigorous setting demonstrate the overwhelming superiority of the proposed framework in both diagnostic accuracy and stability.
- The internal mechanisms of MSFormer are thoroughly analyzed via quantitative confusion matrices and qualitative t-SNE clustering visualizations, with ablation studies further confirming the indispensableness of both the multi scale convolutional branches and Transformer encoders for constructing excellent intra class compactness and inter class separability.

The remainder of this paper is organized as follows: Section 2 details the specific network architecture of the proposed MSFormer, which includes the design mechanism of the multi scale 1D CNN and the internal components of the hierarchical Transformer encoder. Section 3 presents the experimental validation, covering the description of the standard bearing dataset, the selection of baseline models, evaluation metrics, comprehensive diagnosis results, and detailed ablation studies. Section 4 provides a discussion of the advantages and potential limitations of the proposed method. Finally, Section 5 concludes the paper and outlines future research directions.

2. Proposed Method

The proposed MSFormer architecture is an end to end intelligent diagnosis framework. It takes raw 1D vibration signals as input and output the predicted health condition of the machinery. The model comprises three fundamental units: a parallel multi scale feature extraction unit, a hierarchical Transformer encoding unit, and a feature fusion classification head.

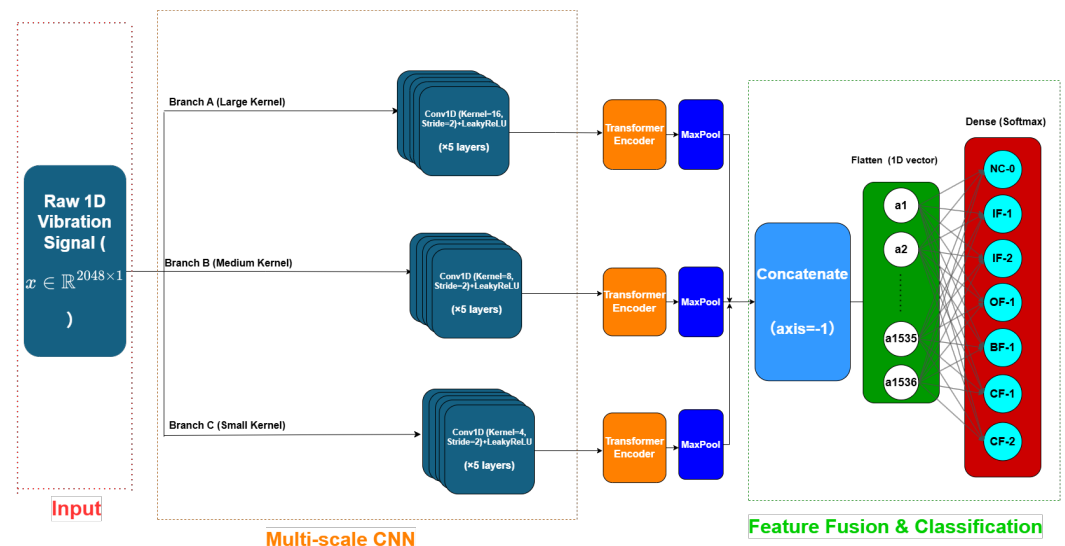


Figure 1. The overall architecture of the proposed MSFormer, comprising the multi scale convolutional feature extraction backbone, the hierarchical Transformer encoder module.

The overall architecture of the proposed MSFormer framework is delineated in Figure 1. Operating as an end to end intelligent diagnostic network, the framework fundamentally comprises three progressive stages: the multi scale convolutional feature extraction backbone, the hierarchical Transformer encoder module, and the diagnostic evaluation block.

Specifically, the raw 1D vibration signal, denoted as $x \in \mathbb{R}^{2048 \times 1}$, is initially fed into the multi scale backbone. To extract multi granular temporal frequency representations under restricted training data regimes, the input is processed simultaneously through three parallel branches configured with varying receptive fields: Branch A ($k = 16$), Branch B ($k = 8$), and Branch C ($k = 4$). This parallel configuration enables the concurrent extraction of macroscopic degradation trends and microscopic localized fault impulses. The extracted multi scale features are subsequently fused via a concatenation layer.

Thereafter, the concatenated representations are propagated into the Transformer encoder module. Within this module, the multi head attention mechanism and feed forward networks (FFN) are leveraged to model long range temporal dependencies and attenuate severe background noise. Ultimately, in the diagnostic phase, the refined global representations are flattened and projected through a fully connected layer to derive the predicted label $\hat{y}$ via the Softmax activation function. To rigorously validate the diagnostic efficacy and physical interpretability of the model, the final network outputs are comprehensively analyzed using a confusion matrix for quantitative classification accuracy and t-SNE visualization for qualitative feature clustering assessment.

2.1. Multi scale Feature Extraction Backbone

Multi scale Feature ExtractionTo capture fault impact characteristics across different frequency resolutions, three parallel convolutional branches (Branch A, B, and C) are utilized. The raw 1D vibration signal, denoted as $X \in \mathbb{R}^{L \times 1}$ (where $L = 2048$ is the data length), is processed by initial convolution layers with varying kernel sizes of 16, 8, and 4, respectively. Large kernels are responsible for global low frequency components, while

small kernels capture high frequency local transients. For a 1D convolutional layer l , the output sequence y^l is computed as:

$$y_i^l = f\left(\sum_{j=1}^K w_j^l \cdot x_{i \times s + j}^{l-1} + b^l\right)$$

where w^l is the convolution kernel of size K , x^{l-1} is the input from the previous layer, s is the stride, and b^l is the bias term. To enhance the non linear expression while preventing the dying ReLU problem, a LeakyReLU activation function is applied after the convolutions:

$$f(x) = \begin{cases} x, & \text{if } x > 0 \\ \alpha x, & \text{if } x \leq 0 \end{cases}$$

where the negative slope coefficient α is set to 0.2. Following the activation, a MaxPooling layer is utilized to downsample the feature maps, reducing computational complexity and extracting dominant features:

$$p_i^l = \max_{j \in W} y_{i \times s + j}^l$$

where W is the pooling window size (set to 2). Each branch consists of 6 stacked Conv1D LeakyReLU MaxPooling blocks, gradually reducing the spatial dimension while increasing the channel depth to 256.

2.2. Transformer Encoding Module

The abstract feature sequences produced by the CNN backbones of the three branches are subsequently mapped into Transformer Encoders. The Transformer models the global dependencies within the sequence through the Multi Head Self Attention (MHSA) mechanism. For the input feature sequence $Z \in \mathbb{R}^{N \times D}$, it is first linearly projected into Queries (Q), Keys (K), and Values (V):

$$Q = ZW_Q, \quad K = ZW_K, \quad V = ZW_V$$

where W_Q, W_K, W_V are learnable weight matrices. The scaled dot product attention is formulated as:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

where d_k is the dimension of the keys. To jointly attend to information from different representation subspaces, MHSA is applied:

$$\text{MHSA}(Z) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h)W_O$$

$$\text{head}_i = \text{Attention}(ZW_{Q_i}, ZW_{K_i}, ZW_{V_i})$$

where $h = 2$ is the number of heads, and W_O is the output projection matrix. The Transformer Encoder incorporates residual connections and Layer Normalization (LN):

$$Z' = \text{LN}(\text{MHSA}(Z) + Z)$$

$$Z_{out} = \text{LN}(\text{FFN}(Z') + Z')$$

where the Feed Forward Network (FFN) consists of two dense layers with intermediate ReLU activation. A dropout rate of 0.1 is adopted to prevent overfitting. Finally, an additional MaxPooling layer is applied to the output of the Transformer Encoder.

2.3. Feature Fusion and Classification

After deep semantic processing, the robust features from Branches A, B, and C are concatenated along the channel dimension to form a comprehensive multi scale global feature representation:

$$F_{fusion} = \text{Concat}(Z_{A_out}, Z_{B_out}, Z_{C_out})$$

This fused feature tensor is flattened into a 1D vector and passed through a fully connected dense layer with a Softmax activation function. The final probability distribution for the $C = 7$ health conditions is computed as:

$$\hat{Y} = \text{Softmax}(W_{cls}F_{flatten} + b_{cls})$$

where W_{cls} and b_{cls} are the weights and biases of the classification layer, respectively.

3. Effectiveness Analysis of the Proposed Method Based on Rolling Bearing Fault Experimental Data

3.1. Experimental and Data Description

To verify the efficiency and strength of the suggested MSFormer, the University of Ottawa (UO) bearing dataset is used in this research [32]. UO data is a well known standard set of data, which set gathered on the basis of a designed rotating machinery test rig, created to emulate the high frequency vibration signals under the complex operating conditions. This experiment has the dataset systematically set to seven different health states to model real world bearing degradation conditions. Such states include a healthy base, and various fault modes with different locations and magnitudes: Normal Condition (NC-0), Inner Race Fault Level 1 (IF-1), *InnerRaceFaultLevel2*(IF-2), Outer Race Fault, Ball Fault, Cage Fault Level 1, and Cage Fault Level 2.

Importantly, the stage of inner ring degradation between IF-1 and IF-2 indicates different levels of crack development with a small width of the defect in the first case and a larger, while more serious structural spall in the second case. Similarly, the degradation profiles of the bearing cage are differentiated under both the influence of the degradation of the bearing cage by the use of CF-1 and CF-2, where the former represents the early stages of local wear or plastic deformation, and the latter represents a more destructive degradation condition, structural fracture or breakage. The multi level severities used can greatly increase the complexity of the diagnostic: faults in the same component have very similar characteristic frequencies, and thus will differ only slightly in their impulse amplitudes and energy distributions, making a rigorous test of the feature decoupling capacity of the diagnostic model.

Moreover, one of the most significant issues related to practical condition monitoring is the extremely low annotation level of fault samples. The training set used in this study is highly limited to ensure rigorous testing of the diagnostic ability and robustness of the model in limited training data conditions, which has 10 samples per class only. The models are trained 100 epochs with mini batch size of 7 during the implementation phase. The parameters of the network are updated via the Adam optimizer, and the initial learning rate of 0.001 is used to ensure effective and consistent convergence.

3.2. Comparative Methods

To comprehensively assess the excellence and strength of the proposed MSFormer, four state of the art deep learning diagnostic models were chosen as comparison means.

- **MCNN:** A classical multiscale CNN lacking global dependency modeling [28].

- **MCSAT:** A multi channel network combined with BiLevelRoutingAttention [29].
- **MRCNN:** A model incorporating residual connections into a multi scale CNN [30].
- **TCAC:** A complex network combining Channel Attention and Transformer modules but lacking shallow multi scale fusion [31].

3.3. Evaluation Metrics

To gain a comprehensive and objective assessment of the performance of the proposed MSFormer in terms of diagnostic accuracy, feature extraction ability, and robustness of the proposed model in complex settings, a joint set of quantitative and qualitative evaluation measures is implemented in this research.

3.3.1. Quantitative Metrics

The performance of the models in terms of overall classification is measured quantitatively with three main measures: Accuracy, Precision, and the F1-Score. To a multi class fault diagnosis task, these measures are derived using the macroscopic aggregation of the True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) of each class:

- **Accuracy:** Represents the proportion of correctly predicted samples out of the total testing samples. It reflects the overall diagnostic correctness of the model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

- **Precision:** Represents the proportion of true fault samples among all samples predicted as a specific fault type. It measures the model's ability to avoid false alarms.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

- **F1-Score:** The harmonic meaning of Precision and Recall. In scenarios where feature boundaries might be blurred, the F1-Score serves as a more comprehensive metric than Accuracy to evaluate the balanced performance of the model.

$$F1-Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (3)$$

- **Robustness Evaluation (Standard Deviation):**

Since the diagnosis is highly sensitive to random data splitting and initialization, relying solely on a single experimental result is scientifically insufficient. Therefore, each model undergoes $J = 10$ independent cross trials. The Standard Deviation (SD) of the evaluation metrics is introduced as a core indicator of model stability and robustness:

$$SD = \sqrt{\frac{1}{J} \sum_{i=1}^J (x_i - \mu)^2} \quad (4)$$

In addition to the aforementioned comprehensive metrics, the confusion matrix is introduced as a detailed quantitative tool to evaluate the classification performance across specific health conditions. By mapping the predicted categories against the true labels, the confusion matrix provides a granular tabular representation of the diagnostic results. The diagonal elements denote the true positive rates (accuracy) for each specific class, while the off diagonal elements explicitly quantify the misclassification probabilities between mutually confusing fault categories. This matrix enables a rigorous assessment of the model's

diagnostic sensitivity, particularly when distinguishing faults with varying severities in the same mechanical component.

3.3.2. Qualitative Metrics

To obtain the intuitive understanding of the information of capability of feature extraction and physical process of the proposed MSFormer, a qualitative evaluation process is taken along with the quantitative measures. In particular, t Distributed Stochastic Neighbor Embedding (t-SNE) algorithm is profiled in this research.

Being one of the most popular non linear dimensionality reduction methods, t-SNE has proven to be very effective to project high dimensional latent representation in a two dimensional visual format that can render the local structure of the data. The high dimensional abstract features produced by the last Flatten layer of the MSFormer are then run through t-SNE to produce a scatter plot during the experimental evaluation phase. This qualitative measure has a clear visualization of the intra class compactness and inter class separability of the trained feature distributions. The t-SNE visualization is a first-hand visual demonstration of the model decoupling of features and resistance to environmental noise, by intuitively showing whether the signals of various health conditions are highly clustered or grossly overlapping.

3.4. Diagnosis Results Analysis

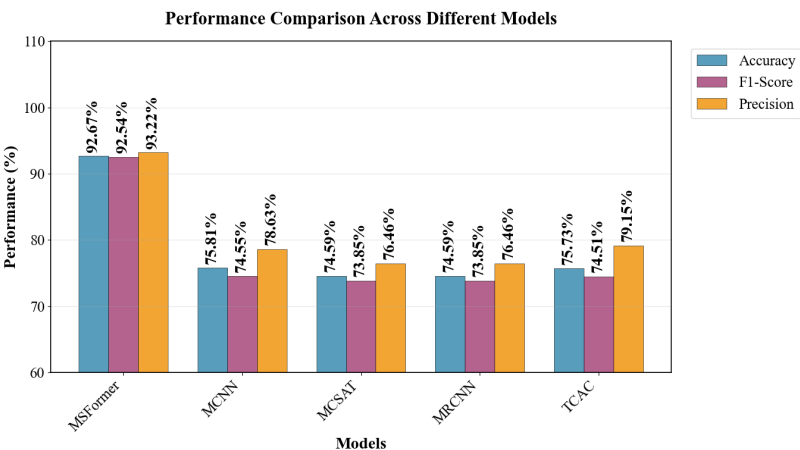


Figure 2. Performance comparison of different diagnostic models. MSFormer demonstrates significant superiority across all three evaluation metrics (Accuracy, F1-score, and Precision) compared to the baseline models (MCNN, MCSAT, MRCNN, and TCAC).

Figure 2 gives a detailed comparison of the performance of each of the five models on three key evaluation measures, namely Accuracy, F1-score, and Precision. The MSFormer, as indicated in the figure, is very effective in terms of performance compared to all other models. The Accuracy of 92.67%, F1-score of 92.54%, and Precision of 93.22% of MSFormer are impressive, given its performance level of more than 92 across all the metrics with only 10 samples per class.

By contrast the comparative models have significantly lower performance, the evaluation metrics of the comparative models are mainly concentrated in the 73%–79% range. For instance, MCNN yields an Accuracy of 75.81% and an F1-score of 74.55%, performing the best among the baselines but still lagging MSFormer by nearly 17%. TCAC achieves an Accuracy of 75.73%, while MCSAT and MRCNN both stall at 74.59%. The overall F1-score and Precision of these baselines remain inferior. The experimental results strongly indicate that relying solely on multi scale CNNs (MCNN, MRCNN) or isolated attention mecha-

nisms (MCSAT, TCAC) is insufficient for extracting discriminative features. MSFormer offers the best trade off, showcasing superior and balanced performance by successfully capturing both local transient impulses and global degradation trends.

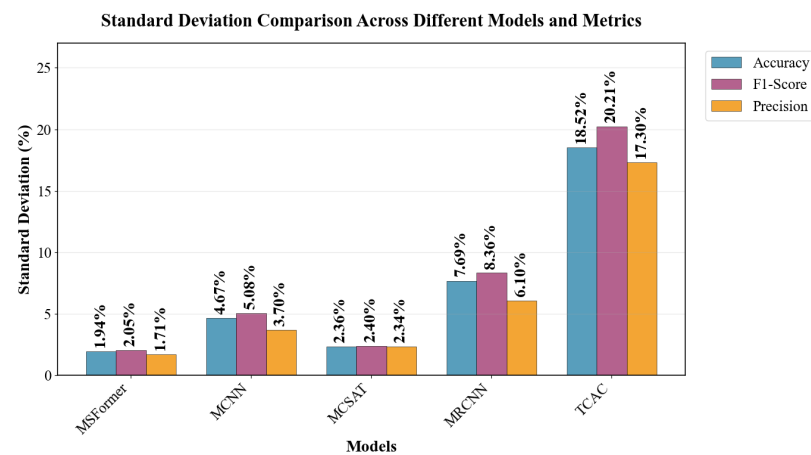


Figure 3. Standard deviation (error rate) comparison of the proposed MSFormer and comparative baseline models. The consistently low standard deviations of MSFormer across all metrics demonstrate its superior stability and robustness.

According to the standard deviation comparison across all evaluation metrics, MSFormer consistently demonstrates the highest stability among all models. As illustrated in Figure 3, its standard deviations in Accuracy, F1-Score, and Precision are 1.94%, 2.05%, and 1.71%, respectively, all remaining well below 2.5%. This indicates that the performance of MSFormer is minimally influenced by data variation and maintains strong robustness.

In comparison, the remaining models exhibit significantly larger fluctuations. TCAC shows extreme instability, with an Accuracy error of 18.52% and an F1-Score error soaring to 20.21%. This suggests that the pure Transformer based structure in TCAC is highly prone to overfitting and severe performance collapse when trained on limited samples without multi scale inductive biases. MRCNN and MCNN also exhibit moderate variance (Accuracy standard deviations of 7.69% and 4.67%, respectively). Although MCSAT achieves a competitive stability (Accuracy error of 2.36%), its overall accuracy is substantially lower than that of MSFormer.

Furthermore, detailed statistical analysis reveals two key findings:

1. The F1 Score exhibits the largest error across almost all models, indicating that this metric is the most sensitive to data variations and class imbalances;
2. Precision consistently shows the smallest error, indicating relatively stable false-positive rates. Taken together, these results show that MSFormer not only surpasses all baseline models in diagnostic accuracy but also provides the most reliable, consistent, and robust predictive behavior;

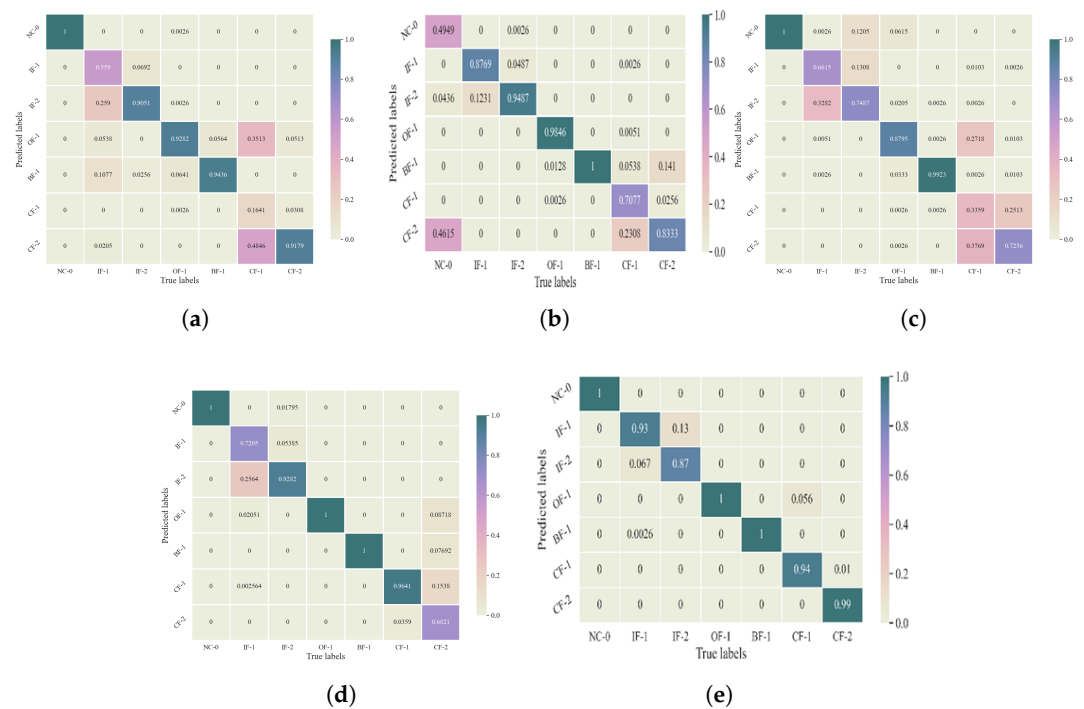


Figure 4. Confusion matrices of the five diagnostic models: (a) MCNN, (b) MCSAT, (c) MRCNN, (d) TCAC, and (e) MSFormer. While the baseline models struggle to distinguish similar fault signatures (e.g., IF-1 and IF-2), the MSFormer achieves the clearest diagonal matrix and significantly mitigates mutual misclassifications, demonstrating its superior fine-grained fault recognition capability.

By comparing the confusion matrices shown from (a) to (e), the fine grained classification capabilities of the models can be observed. The research found that compared with MSFormer, the baseline models show high rates of mutual misidentification, particularly among fault types with similar physical manifestations.

For instance, the MCNN and MRCNN models show a high rate of misclassifying IF-1 (Inner Race Fault 1) as IF-2 and easily confuse CF-1 with CF-2 (Cage Faults). TCAC exhibits the most severe confusion, almost completely failing to distinguish between CF-1 and CF-2 and mistakenly identifying early stage inner race faults as outer race faults. The MCSAT model performs better than TCAC but still leaves a noticeable gap in classifying IF-2.

Conversely, the experimental results demonstrate that MSFormer achieves the clearest diagonal matrix. It yields a 100% recognition rate for NC-0, OF-1, and BF-1. More importantly, it significantly suppresses the misclassification between IF-1 and IF-2, achieving recognition rates of 0.93 and 0.87, respectively. This demonstrates that MSFormer possesses superior recognition capability for subtle and overlapping fault signatures compared to the other four models.

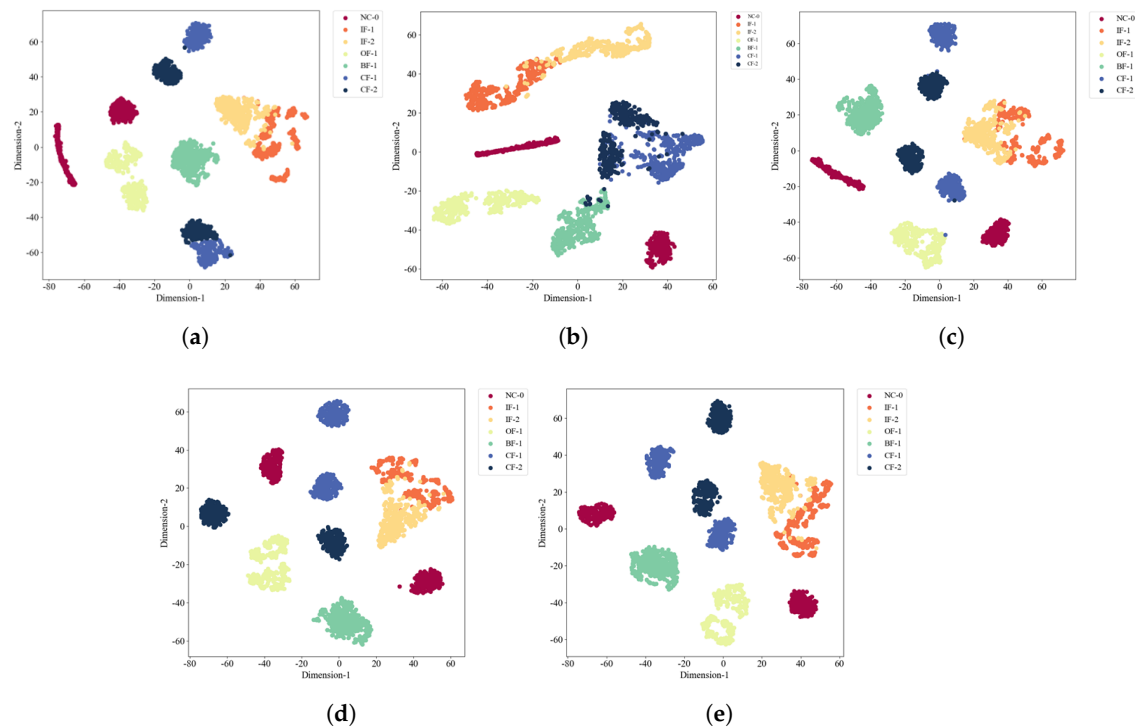


Figure 5. The t-SNE feature visualization of the five diagnostic models: (a) MCNN, (b) MCSAT, (c) MRCNN, (d) TCAC, and (e) MSFormer. Compared to the baseline models, MSFormer exhibits the optimal clustering performance, achieving the highest intra class compactness and the clearest inter class separability.

The t-SNE was used to reduce the dimensionality of high dimensional vibration features generated by the final layer of each model into a two dimensional space. Combined with the visualization results, the analysis is as follows:

MCNN and MCSAT: Severe feature confusion can be observed. The boundaries between IF-1 and IF-2, CF-1 and CF-2 are completely blurred and scattered. This demonstrates that without an effective multi scale feature extraction backbone, these models fail to project samples into a separable feature space.

TCAC and MRCNN: The clustering effect has improved. The inter class separation is somewhat enhanced, and clusters for clear faults like NC-0 and BF-1 are relatively compact. However, partial overlap still exists between IF-2 and IF-1, indicating that isolated CNN mechanisms still lack the long range dependency modeling capability to distinguish subtle inner race fault differences.

MSFormer: The clustering effect is optimal. The 7 fault categories almost form compact and tightly bound clusters. Apart from a very slight boundary proximity between IF-1 and IF-2, all other classes are completely separated with large inter class margins. This visually verifies that the synergy of the multi scale CNN structure and the Transformer module effectively improves the class discriminability of features, providing strong support for high diagnostic accuracy.

3.5. Ablation Experiment

To analyze the contribution of each key component in the proposed method to performance improvement, ablation experiments were conducted. The comparative methods were constructed as follows.

- **Model 1;**

1.

Model1-A: This model retains only the Transformer module and a single branch A;

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2.

Model1-B: This model retains only the Transformer module and a single branch B;

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3.

Model1-C: This model retains only the Transformer module and a single branch C.

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- **Model 2:** This model is a multi scale CNN model without the Transformer module;

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- **Model 3:** Three models with a CNN model without multi scale functionality.

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1.

Model3-A: This model only remains branch A;

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2.

Model3-B: This model only remains branch B;

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3.

Model3-C: This model only remains branch C.

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- **MSFormer.**

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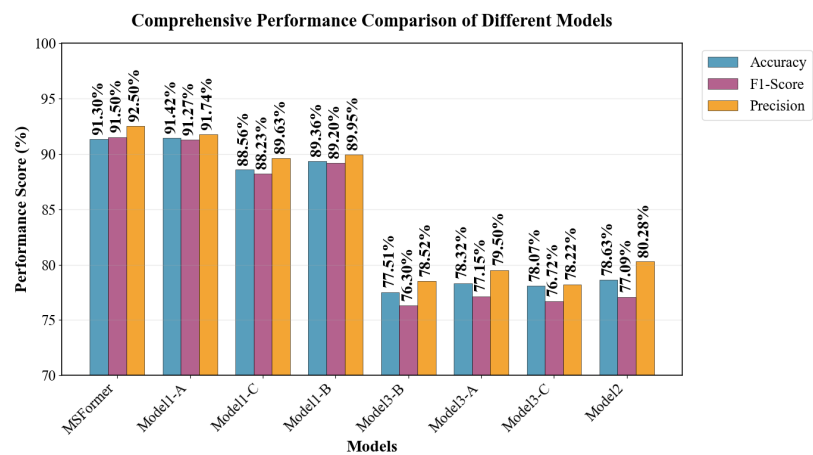


Figure 6. Comprehensive performance comparison of MSFormer and comparative baseline models. The results demonstrate that MSFormer achieves the highest overall scores across Accuracy, F1-Score, and Precision, highlighting its superior fault diagnostic capability.

Figure 6 presents a comprehensive comparison of the performance of all models across four evaluation metrics, which include Accuracy, F1-score, Precision, and Recall. As shown in the figure, MSFormer and Model1 series (Model1-A, Model1-C, and Model1-B) outperform the other models. Among them, MSFormer and Model1-A even achieve performance levels above 0.90 on all four metrics. However, MSFormer yields the best overall results with the highest Precision and competitive Accuracy and Recall, which is better than model1-A. The average of four metrics about MSFormer is 91.65%, which is slightly better than Model1-A, which is 91.4%.

In contrast, the Model3 series (Model3-A, Model3-B, and Model3-C) exhibits relatively lower performance across all metrics, with Accuracy and F1-scores primarily in the 76%–79%. Although Model3-A shows slightly improved Precision (79.5%), the overall F1-score and Recall remain inferior to those of the top performing models.

Model2 achieves moderate performance, with most metrics around 78%, and exhibits a noticeable gap compared to the MSFormer and Model1 series. While Model2 demonstrates a relatively higher Precision (80.3%), its Accuracy and F1-score remain substantially lower than those of the leading models, indicating that it tends to generate more false positives.

Overall, the experimental results indicate that MSFormer and Model1 based variants offer the best trade off among Accuracy, Precision, Recall, and F1, showcasing superior and balanced performance. Meanwhile, the Model3 series and Model2 lag in all major evaluation indicators, highlighting the effectiveness of the MSFormer and Model1 architectures for this classification task

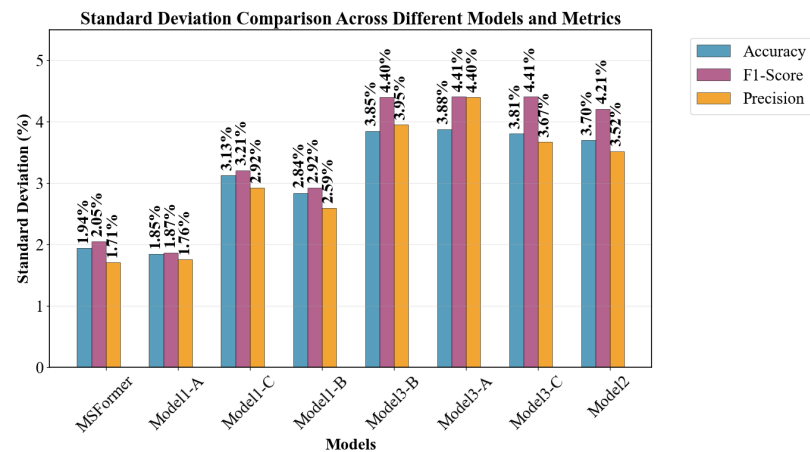


Figure 7. Standard deviation comparison of different models across evaluation metrics (Accuracy, F1-Score, and Precision). MSFormer exhibits the lowest standard deviations across all metrics, confirming its superior stability and robustness against data variations compared to other models.

According to the standard deviation comparison across all evaluation metrics, MSFormer consistently demonstrates the highest stability among all models. Its standard deviations in Accuracy, F1-Score, and Precision are the lowest in the entire group, all remaining close to or below approximately 2.0%. This indicates that the performance of MSFormer is minimally influenced by data variation and maintains strong robustness.

In comparison, the remaining models—including Model1-A, Model1-C, Model1-B, Model3-B, Model3-A, Model3-C, and Model2—exhibit noticeably larger fluctuations. For instance, models in the Model3 family show the highest variability, with standard deviations reaching 4.0%–4.4% across the metrics, reflecting significant instability. Even relatively better performing models such as Model1-A and Model1-B still show higher variance than MSFormer, with standard deviations generally ranging from 1.7% to 3.0%.

Taken together, these results show that MSFormer not only surpasses Model1-A but also outperforms all other models in terms of stability. Its substantially lower standard deviations across all metrics confirm that MSFormer provides the most reliable, consistent, and robust predictive behavior among the compared models. This makes MSFormer the most stable and practically deployable model in this study.

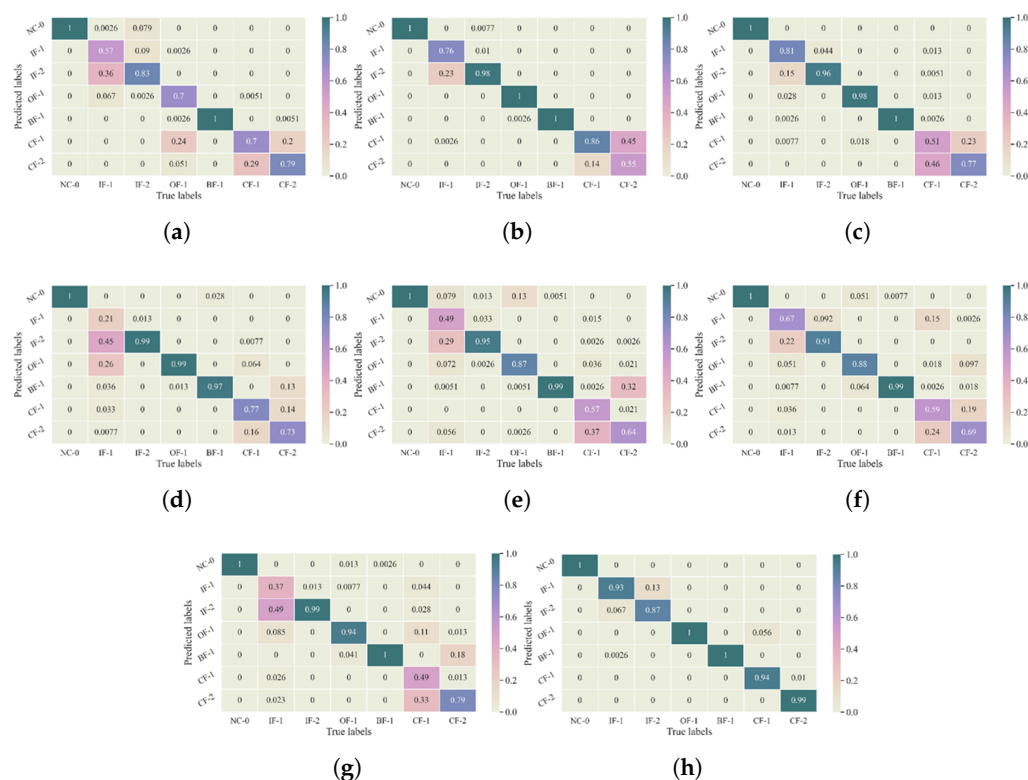


Figure 8. Confusion matrices of the four diagnostic models: (a) Model1-A, (b) Model1-B, (c) Model1-C, (d) Model2, (e) Model3-A, (f) Model3-B, (g) Model3-C, and (h) MSFormer.

By comparing the confusion matrices, research found that compared with MSFormer, Model 1 shows an extremely high rate of mutual misidentification between CF-2 and CF-1. Additionally, a relatively high rate of misclassifying IF-1 as IF-2 also exists. Model 2 tends to misclassify IF-1 as IF-2 and OF-2, which also easily confuses CF-1 with CF-2. Model 3 shows a tendency to misclassify IF-1 as IF-2 and CF-1 as CF-2. Moreover, it has low recognition capability for both CF-1 and CF-2. The experimental results demonstrate that MSFormer shows better recognition than the other three models.

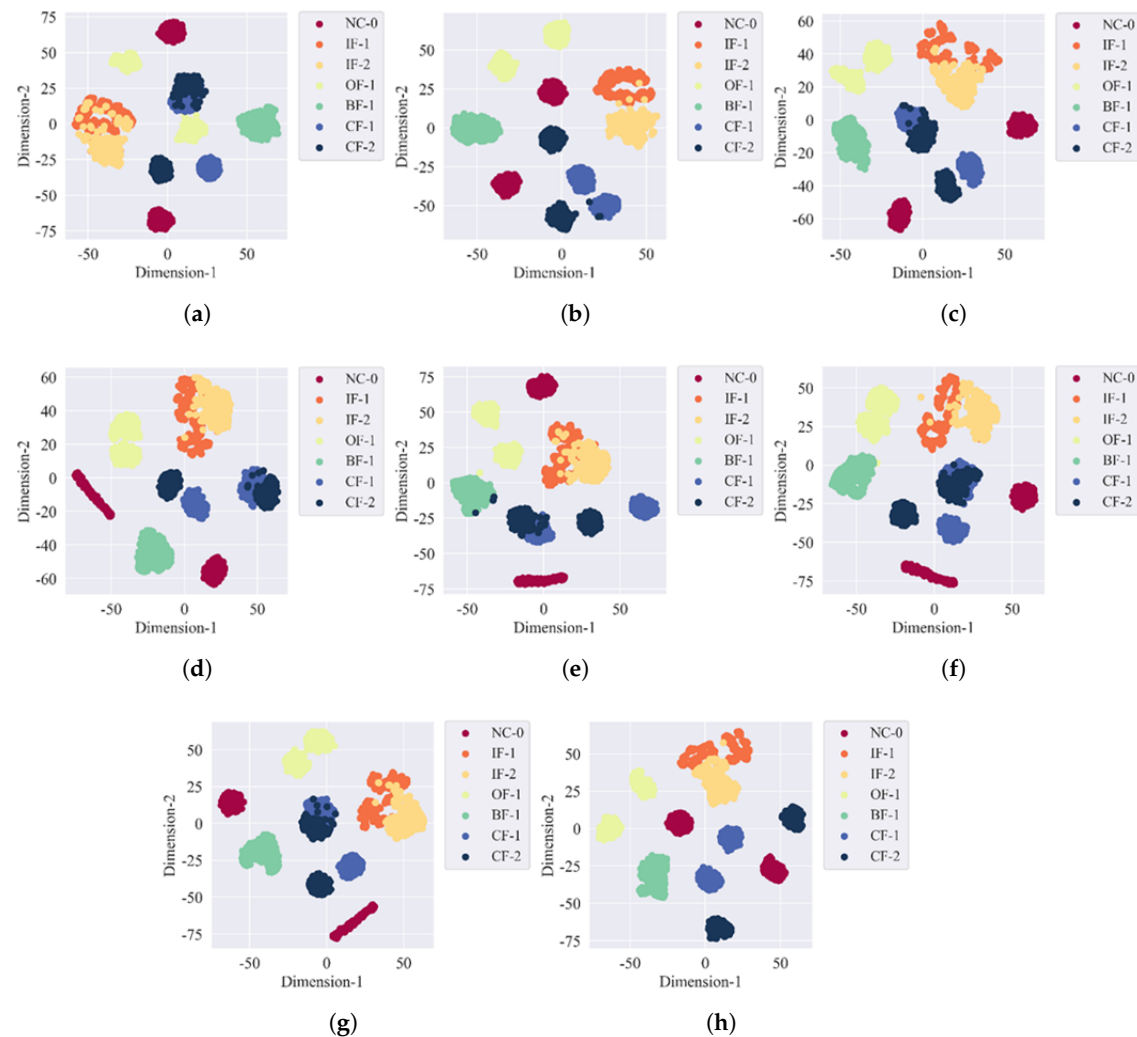


Figure 9. The t-SNE feature visualization of the four diagnostic models: (a) Model1-A, (b) Model1-B, (c) Model1-C, (d) Model2, (e) Model3-A, (f) Model3-B, (g) Model3-C and (h) MSFormer.

t-SNE was used to reduce the dimensionality of high dimensional vibration features to a two dimensional space. Combined with the visualization results in the document, the analysis is as follows.

- Model 3 (Basic CNN):** Severe feature confusion can be observed. The feature points of CF1-1 and CF-2 overlapped significantly. The boundaries between IF-1 and IF-2 are blurred and scattered. This demonstrates that the single branch CNN can only extract single scale features, making it unable to distinguish weak local features of incipient faults;
- Model 1 (Single Branch + Transformer):** The clustering effect was improved. Clusters of NC-0 and BF-1 were relatively compact, but partial overlap still existed between IF-2 and IF-1. This indicates that although the Transformer module can focus on key features, the single branch fails to cover multi scale fault features, leading to confusion between similar fault categories;
- Model 2 (Multi Branch + No Transformer):** The inter class separation was significantly enhanced. The boundaries of clusters for the 7 fault categories were relatively clear, with only slight overlap between CF-2 and CF-1. But partial overlap still existed between IF-2 and IF-1. This confirms that the multi scale structure can provide rich feature dimensions; however, without the long range dependency modeling capability

of the Transformer module, it still cannot distinguish the subtle differences between cage faults;

- **MSFormer** The clustering effect was optimal. The 7 fault categories almost formed compact and non overlapping clusters. Moreover, the other classes are completely separated except for IF-1 and IF-2 overlapping. But there is also less overlap than other models. This verifies that the synergy of the multi scale structure and Transformer module can effectively improve the class discriminability of features, providing support for high diagnostic accuracy.

4. Discussion

The empirical results demonstrate that the proposed MSFormer exhibits distinctive merits in the diagnosis of mechanical faults under complex operating conditions. Primarily, the parallel multi scale CNN backbone effectively isolates multi granular temporal frequency features from vibration signals, while the hierarchical Transformer encoder captures comprehensive global representation patterns through long range dependency modeling. Furthermore, the framework maintains an exceptional generalization capacity under restricted data regimes, ensuring elevated diagnostic accuracy with only 10 training samples per category and effectively mitigating the risk of overfitting. Additionally, qualitative t-SNE visualizations confirm the model's heightened sensitivity to subtle fault severities, successfully decoupling overlapping degradation stages (e.g., IF-1 versus IF-2) within identical mechanical components.

Despite its pronounced diagnostic efficacy, certain limitations warrant further investigation. First, this study did not conduct quantitative Signal to Noise Ratio (SNR) experiments involving artificial noise injection; therefore, rigorously verifying the model's robustness under severe environmental noise fluctuations remains a critical priority for future work. Furthermore, compared to purely lightweight CNNs, the self attention mechanisms inherent in the model introduce higher computational overhead and memory allocation. Consequently, future research must explore model compression techniques to facilitate deployment on edge devices.

5. Conclusions

This paper introduces a new, powerful mechanism for a fault diagnosis algorithm called MSFormer to address the essential issues of data shortage. The model provides an ideal combination of local impulse extraction and global dependency modeling by engaging parallel multi scale convolutional branches and hierarchical Transformer encoders innovatively. When tested on the UO dataset using a highly constrained training set (10 samples per class), MSFormer exhibits a higher accuracy of 92.67, which is much higher than classical baseline models. Also, its exceptional robustness is systematically confirmed by rigorous stability experiments. Future research will focus on the optimization of self attention computational complexity to enable it to operate on edge computing devices.

References

1. Neupane, D.; Bouadjenek, M. R.; Dazeley, R.; Aryal, S. Data-driven machinery fault diagnosis: A comprehensive review. *Neurocomputing* **2025**, *627*, 129588. <https://doi.org/10.1016/j.neucom.2025.129588>.
2. Lei, Y.; Jia, F.; Lin, J.; Xing, S.; Ding, S. An intelligent fault diagnosis method using unsupervised feature learning towards mechanical big data. *IEEE Transactions on Industrial Electronics* **2016**, *63*, 3137–3147. <https://doi.org/10.1109/TIE.2016.2519325>.
3. Jia, F.; Lei, Y.; Lin, J.; Zhou, X.; Lu, N. Deep neural networks: A promising tool for fault characteristic mining and intelligent diagnosis of rotating machinery with massive data. *Mechanical Systems and Signal Processing* **2016**, *72*, 303–315.
4. Chen, G.; et al. Enhancing Reliability Through Interpretability: A Comprehensive Survey of Interpretable Intelligent Fault Diagnosis in Rotating Machinery. *IEEE Access* **2024**, *12*, 103348–103379. <https://doi.org/10.1109/ACCESS.2024.3430010>.

5. Smith, W. A.; Randall, R. B. Rolling element bearing diagnostics using the Case Western Reserve University data: A benchmark study. *Mechanical Systems and Signal Processing* **2015**, *64*, 100–131. 438
6. Wang, H.; et al. A reliable feature-assisted contrastive generalization net for intelligent fault diagnosis under unseen machines and working conditions. *Mechanical Systems and Signal Processing* **2023**, *188*, 110003. 439
7. Li, S.; Liu, J.; Ma, H.; Yang, N. Multimodal Adaptive Signal Fusion for Domain Generalization in Imbalanced Few-Shot Rotating Machinery Fault Diagnosis. *IEEE Transactions on Instrumentation and Measurement* **2025**, *74*, 1–16. 440
<https://doi.org/10.1109/TIM.2025.3643024>. 441
8. Lu, Z.; Liang, L.; Zhu, J.; Zou, W.; Mao, L. Rotating Machinery Fault Diagnosis Under Multiple Working Conditions via a Time-Series Transformer Enhanced by Convolutional Neural Network. *IEEE Transactions on Instrumentation and Measurement* **2023**, *72*, 1–11. <https://doi.org/10.1109/TIM.2023.3318707>. 442
9. Shen, H.; Tang, X.; Luo, Y.; Xie, F.; Shi, Z. Online Open-Circuit Fault Diagnosis for Neutral Point Clamped Inverter Based on an Improved Convolutional Neural Network and Sample Amplification Method Under Varying Operating Conditions. *IEEE Transactions on Instrumentation and Measurement* **2024**, *73*, 1–12. <https://doi.org/10.1109/TIM.2023.3348886>. 443
10. Zhang, Y.; Ding, J.; Li, Y.; Ren, Z.; Feng, K. Multi-modal data cross-domain fusion network for gearbox fault diagnosis under variable operating conditions. *Engineering Applications of Artificial Intelligence* **2024**, *133*, 108236. 444
<https://doi.org/10.1016/j.engappai.2024.108236>. 445
11. Zhu, X.; Zhao, X.; Yao, J.; Deng, W.; Shao, H.; Liu, Z. Adaptive Multiscale Convolution Manifold Embedding Networks for Intelligent Fault Diagnosis of Servo Motor-Cylindrical Rolling Bearing Under Variable Working Conditions. *IEEE/ASME Transactions on Mechatronics* **2024**, *29*, 2230–2240. <https://doi.org/10.1109/TMECH.2023.3314215>. 446
12. Zhao, X.; Yao, J.; Deng, W.; Ding, P.; Zhuang, J.; Liu, Z. Multiscale Deep Graph Convolutional Networks for Intelligent Fault Diagnosis of Rotor-Bearing System Under Fluctuating Working Conditions. *IEEE Transactions on Industrial Informatics* **2023**, *19*, 166–176. <https://doi.org/10.1109/TII.2022.3161674>. 447
13. Liu, Z.; Peng, Z. Few-shot bearing fault diagnosis by semi-supervised meta-learning with graph convolutional neural network under variable working conditions. *Measurement* **2025**, *240*, 115402. <https://doi.org/10.1016/j.measurement.2024.115402>. 448
14. Song, B.; Liu, Y.; Fang, J.; Liu, W.; Zhong, M.; Liu, X. An optimized CNN-BiLSTM network for bearing fault diagnosis under multiple working conditions with limited training samples. *Neurocomputing* **2024**, *574*, 127284. 449
<https://doi.org/10.1016/j.neucom.2024.127284>. 450
15. Wang, X.; Mao, D.; Li, X. Bearing fault diagnosis based on vibro-acoustic data fusion and 1D-CNN network. *Measurement* **2021**, *173*, 108518. <https://doi.org/10.1016/j.measurement.2020.108518>. 451
16. Chang, X.; Lyu, J.; Ma, H. Motor Bearing Fault Diagnosis Based on EMD-CNN-Transformer. In *The Proceedings of the 20th Annual Conference of China Electrotechnical Society; Lecture Notes in Electrical Engineering*, Vol. 1559; Springer: Singapore, **2026**. 452
https://doi.org/10.1007/978-981-95-7576-3_6 453
17. Zhang, Y.; Zhou, T.; Huang, X.; Cao, L.; Zhou, Q. Fault diagnosis of rotating machinery based on recurrent neural networks. *Measurement* **2021**, *171*, 108774. <https://doi.org/10.1016/j.measurement.2020.108774>. 454
18. Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; Polosukhin, I. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*; **2017**, *30*. 455
19. Dosovitskiy, A.; Beyer, L.; Kolesnikov, A.; Weissenborn, D.; Zhai, X.; Unterthiner, T.; Dehghani, M.; Minderer, M.; Heigold, G.; Gelly, S.; Uszkoreit, J.; Houlsby, N. An image is worth 16x16 words: Transformers for image recognition at scale. In *Proceedings of the International Conference on Learning Representations (ICLR)*; **2021**. 456
20. Zhou, H.; Zhang, S.; Peng, J.; Zhang, S.; Li, J.; Xiong, H.; Zhang, W. Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, **2021**, *35*(12), 11106–11115. 457
<https://doi.org/10.1609/aaai.v35i12.17325>. 458
21. Ding, Y.; Jia, M.; Miao, Q.; Cao, Y. A novel time–frequency Transformer based on self–attention mechanism and its application in fault diagnosis of rolling bearings. *Mechanical Systems and Signal Processing* **2022**, *168*, 108616. 459
<https://doi.org/10.1016/j.ymssp.2021.108616>. 460
22. An, Y.; Zhang, K.; Chai, Y.; Zhu, Z.; Liu, Q. Gaussian Mixture Variational-Based Transformer Domain Adaptation Fault Diagnosis Method and Its Application in Bearing Fault Diagnosis. *IEEE Transactions on Industrial Informatics* **2024**, *20*, 615–625. 461
<https://doi.org/10.1109/TII.2023.3268750>. 462
23. Li, S.; Yu, J. A Multisource Domain Adaptation Network for Process Fault Diagnosis Under Different Working Conditions. *IEEE Transactions on Industrial Electronics* **2023**, *70*(6), 6272–6283. <https://doi.org/10.1109/TIE.2022.3194654>. 463
24. Zhao, C.; Shen, W. Mutual-assistance semisupervised domain generalization network for intelligent fault diagnosis under unseen working conditions. *Mechanical Systems and Signal Processing* **2023**, *189*, 110074. <https://doi.org/10.1016/j.ymssp.2022.110074>. 464
25. Yang, X.; Yuan, X.; Ye, T.; Zhu, W.; Zhou, F.; Jin, J. PSNN-TADA: Prototype and Stochastic Neural Network-Based Twice Adversarial Domain Adaptation for Fault Diagnosis Under Varying Working Conditions. *IEEE Transactions on Instrumentation and Measurement* **2024**, *73*, 1–12, Art no. 3500212. <https://doi.org/10.1109/TIM.2023.3338678>. 465

26. An, Z.; Jiang, X.; Liu, J. Mode-Decoupling Auto-Encoder for Machinery Fault Diagnosis Under Unknown Working Conditions. *IEEE Transactions on Industrial Informatics* **2024**, *20*(3), 4990–5003. <https://doi.org/10.1109/TII.2023.3331129>. 493
27. Chen, X.; Zhang, B.; Gao, D. Bearing fault diagnosis base on multi-scale CNN and LSTM model. *Journal of Intelligent Manufacturing* **2021**, *32*, 971–987. <https://doi.org/10.1007/s10845-020-01600-2>. 494
28. Xia, Y.; Zhang, D.; Liu, C.; Cao, Z.; Su, Y.; Chen, Y. Fault diagnosis of air handling units based on an MCNN-Transformer ensemble learning. *Journal of Process Control* **2025**, *154*, 103526. <https://doi.org/10.1016/j.jprocont.2025.103526>. 495
29. Zhuang, X.; Ge, J.; Mao, X.; Zhou, D.; Yao, H.; Sun, W.; Li, L.; Xiang, J. A novel lightweight model combined with convolutional neural network and transformer for gearbox fault diagnosis using infrared thermal images. *Engineering Applications of Artificial Intelligence* **2025**, *162*, 112704. <https://doi.org/10.1016/j.engappai.2025.112704>. 496
30. Yao, Z.; Jiang, Q.; Gu, X. A fusion model fault diagnosis scheme based on multibranch multiscale residual convolutional network with Transformer. *Chemical Engineering Research and Design* **2025**, *220*, 667–681. <https://doi.org/10.1016/j.cherd.2025.07.042>. 497
31. Zhang, J.; Zhang, M.; Wang, D.; Yang, M.; Liang, C. Multi-scale convolutional sparse attention transformer: A lightweight fault diagnosis model for rotating machinery. *Neurocomputing* **2025**, *650*, 130934. <https://doi.org/10.1016/j.neucom.2025.130934>. 498
32. Huang, H.; Baddour, N. Distributed hidden Markov model for machine health assessment. *IEEE Transactions on Instrumentation and Measurement* **2010**, *59*, 3179–3189. <https://doi.org/10.1109/TIM.2010.2047306>. Dataset available at: <https://data.mendeley.com/datasets/v43hmbwaxpm/2>. 499
33. Zhang, T.; Chen, J.; Ye, Z.; Liu, W.; Tang, J. Prior knowledge-informed multi-task dynamic learning for few-shot machinery fault diagnosis. *Expert Systems with Applications* **2025**, *271*, 126439. <https://doi.org/10.1016/j.eswa.2025.126439>. 500
34. Zhang, T.; Chen, J.; He, S.; Zhou, Z. Prior knowledge-augmented self-supervised feature learning for few-shot intelligent fault diagnosis of machines. *IEEE Transactions on Industrial Electronics* **2022**, *69*, 10573–10584. <https://doi.org/10.1109/TIE.2022.3140403>. 501